unlike conventional capacitors that use a solid and dry dielectric material, such as teflon, polyethylene and paper, ultracapacitors use a liquid or wet electrolyte between their electrodes. A supercapacitor or electric double-layer capacitor, as shown in Fig. 3, utilises large-surface area electrodes and thin electrolytic dielectrics to achieve capacitance values greater than any other capacitor type available today. Energy storage in capacitors is by means of static charge and does not involve the electro-chemical process inherent to batteries. These store charge as a coating of ions adsorbed on the electrodes' surface. Ions are separated from the electrolyte by a charging current, and propelled toward their respective electrodes. The membrane serves to separate the ions, so that a net charge separation can be maintained. This dual surface layer is called electric double layer.

Supercapacitors have separation of charge at an electric field which is only fractions of a nanometre, compared to micrometres for most polymer film capacitors. In doing so, supercapacitors are able to attain greater energy densities while still maintaining the characteristic of high power density of conventional capacitors. These are therefore more of an electrochemical device similar to an electrolytic capacitor. They store charge through reversible ion adsorption at the surface of high-surface-area carbon. Different materials, such as various carbon materials, mixed-metal oxides and

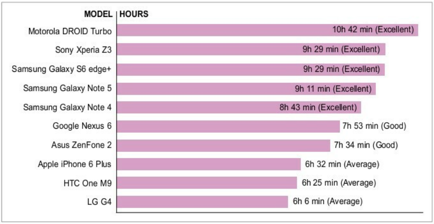

Fig. 2: Battery life of mobile phones

conducting polymers, have been used for supercapacitor electrodes. Advances in carbon-based materials, namely, graphene, have increased their energy density to almost the level of batteries.

Graphene is a thin layer of pure carbon, tightly packed and bonded together in a hexagonal honeycomb lattice. It is widely regarded as the most suitable material for supercapacitors as it is the thinnest compound known to man at one atom thickness, as well as the best known conductor. It also has amazing strength and light absorption characteristics, and is even considered eco-friendly and sustainable, as carbon is widespread in nature and is a part of the human body.

Graphene-based materials are considered as one of the most promising materials for the next-generation, flexible, thin-film supercapacitors due to their unique structure and properties: Their two-dimensional structure can provide a large surface area, which serves as an extensive transport platform for electrolytes. The high conductivity of graphene sheets enables a low diffusion resistance, leading to enhanced power and energy density. The superior mechanical property allows graphene sheets to be easily assembled into free-standing films with robust mechanical stability. Among graphene-based 2D films, graphene papers have attracted much attention due to their tunable thickness.

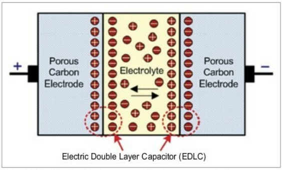

Fig. 3: Electric double-layer capacitor (supercapacitor)

But as devices shrink, integrating the storage element as close as possible to the electronic circuit (directly on a chip) is another challenge. Electrochemical capacitors with a high surface-to-volume ratio of the active material have high energy and power densities; this is further enhanced in micro-supercapacitors.

Porous activated, templated and carbide-derived carbons, multi- and single-walled carbon nanotubes, onion-like carbon (OLC) and multilayer graphene have been used as electrode materials in supercapacitors. Progress in micro-fabrication technology has enabled on-chip micro-supercapacitors in an interdigitated planar form in contrast to the conventional sandwich structure. The planar form is compatible with integrated circuits. Micro-supercapacitors, besides their miniaturised structure, have high power density, high rate capability and high frequency response, which are crucial for most applications.

Among all the options available, only onion-like carbon has produced micro-supercapacitors capable of